MATH 323: Probability

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Abstract

Contents

1 Introduction **1**

2 Prerequisite Knowledge **1**

3 Appendix **4**

4 Solutions **4**

5 Useful Links **4**

1 Introduction

These notes were written for MATH 323 (Probability) at McGill University, Summer 2026. The course is taught by Magid Sabbagh and meets in Steward Biology Building S1/3 (S1/3) every Monday, Tuesday, Wednesday, and Thursday from 8:35 am to 10:55 am.

Grading scheme:

$$20\% \text{ assignments} + 5\%\text{Multiple Choice Quiz} + 25\% \text{ midterm} + 50\% \text{ final} \quad (1)$$
$$\text{or} \quad (2)$$
$$20\% \text{ assignments} + 5\%\text{Multiple Choice Quiz} + 75\% \text{ final} \quad (3)$$

whichever is higher. The book used for this class is: "Mathematical statistics with application, 7th edition." Midterm will be an 1h30 long and will have all 2h30 hours to complete it in class.

2 Prerequisite Knowledge

May 4, 2026

Basic Set Theory:

- A set is a well defined collection of objects, defined in a way such that we can determine wether or not an element x belongs to the set.
- If A is a set, we say that $a \in A$, if a is an element of A.
- If A is a set, we say that $a \notin A$ if a is not an element of A.
- We specify a set either by listing its elements, if possible on by a property that defines the set.
- In general, we will be working in a universal set Ω.
- Universal set is a set in which all elements belong, like the complex set.
- Operations on sets, let Ω be a universal set, $A, B \subseteq \Omega$. Then we can define the following operations on sets:
  - Subset: $A \subseteq B$ if $x \in A$ implies $x \in B$.
  - Equality: $A = B$ if $A \subseteq B$ and $B \subseteq A$.
  - Proper subset: $A \subsetneq B$ if $A \subseteq B$ and $A \neq B$.
  - Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
  - Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
  - Difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$
  - Complement: $A^c = \{x : x \in \Omega \text{ and } x \notin A\}$

– Singleton: $\{x\} = \{y : y = x\}$, the set that contains only the element x.

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Example. *1) Let Ω be a set. $\mathbb{F} = \{\phi, \Omega\}$ is a σ-algebra on Ω.*
2) Let Ω be a set. Then the set of all subset of Ω is a σ-algebra on Ω. (The power set of Ω denoted by Z^{Ω}).
3) Let ω be a set. Define $\mathbb{F} = \{A \subseteq \Omega . A$ is countable on $A^c$is countable$\}$. Then $\mathbb{F}$ is a σ-algebra on Ω. 4) Suppose $A \subseteq \Omega$. $\mathbb{F} = \{\phi, A, A^c, \Omega\}$ is a σ-algebra.

Example. *Flip a coin. $\Omega = \{H, T\}$. $2^n = \{\phi, \{H\}, \{T\}, \{H, T\}\}$.*

Definition 1. *Let Ω be the sample space (...) to an experiment and $\mathbb{F}$ a σ-algebra of subsets of Ω. (...)*

1. *$P(A) \geq 0$ for all $A \in \mathbb{F}$.*
2. *$P(\Omega) = 1$.*
3. *If $A_1, A_2, \ldots$ is a collection of pairwise disjoint subsets of Ω that belong to F then are sets in $\mathbb{F}$ such that $A_i \subseteq A_{i+1}$ for all i, then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. (Pairwise disjoint $A_k \cap A_l = \phi$ if $k \neq l$)*

Properties:

1. *If $A \subseteq B$ then $P(A) \leq P(B)$. Proof: $B = A \cup (B \setminus A)$ and $P(B) = P(A) + P(B \setminus A)$ since A and $B \setminus A$ are disjoint. $P(B) = P(A) + P(B \setminus A) \geq P(A)$ since $P(B \setminus A) \geq 0$.*

Corollary 0.1. *Let $A \in \mathbb{F}$. (A is an event). Then $P(A) \leq 1$. Proof: $A \subseteq \Omega$, (...) by the previous proof, $P(A) \leq P(\Omega) = 1$.*

Proposition 1. *$P(A^c) = 1 - P(A)$. Proof: $\Omega = A \cup A^c$ and $A \cap A^c = \phi$. $P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) = 1$, so $P(A^c) = 1 - P(A)$.*

Let A, B be events $(A, B \in \mathbb{F})$. Then $A \cap B \in \mathbb{F}$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $A \cup B = A \cup (B \cap A^c)$, $A \cap (B \cap A^c) = \phi$, $P(A \cup B) = P(A) + P(B \cap A^c)$.

Lemma 1. *$P(B) = P(A^c \cap B) + P(A \cap B)$ and $P(A \cap B) = P(B) - P(A^c \cap B)$, $B = (A \cap B) \cup (A^c \cap B)$, $A \cap B$, $A^c \cap B$ are disjoint $P(B) = P(A \cap B) + P(A^c \cap B)$ and hence $P(A \cap B) = P(B) - P(A \cap B)$ and hence $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.*

The axioms define a probability, require THE countable additivity for piecewise disjoint sets IF $\{A\}_{i=1}^{\infty}$ are a collection of events such that $A_k \cap A_l = \phi$ if $k \neq l$ then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. If $\{A\}_{i=1}^{\infty}$ are not disjoint, how can we relate $P(\cup_{i=1}^{\infty} A_i)$ and $\sum_{i=1}^{\infty} P(A_i)$?

Proposition 2. *Let $\{A\}_{i=1}^{\infty}$ be a collection of events. Then $P(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i)$. Proof: define $A_i^{\star} = A_i$ and $A_2^{\star} = A_2 \setminus A_1$, $A_3^{\star} = A_3 \setminus (A_1 \cup A_2)$, $\ldots$, $A_n^{\star} = A_n \setminus (\cup_{i=1}^{n-1} A_i)$, $\ldots$, for $n \geq 2$.*

First, $\{A\}_{i=1}^{\infty}$ are pairwise disjoint. To show this, let $k \neq l$ be positive integers. We need to show that $A_k^\star \cap A_l^\star = \phi$. Assume that $l > k$ and suppose$A_k^\star \cap A_l^\star \neq \phi$. Let $x \in A_k^\star \cap A_l^\star$. Then $x \in A_l^\star$ and $x \in A_k^\star$. Since $x \in A_l^\star$ and $x \in A_k^\star$. Since $x \in A_l^\star$ then $x \in A_l$ but $x \notin \cup_{i=1}^{l-1} A_i$. Since $k < l$ then $x \notin A_k$. Futhermore, since $x \in A_k^\star$, then $x \in A_k$, we have here a contradiction. Hence $\{A_i^\star\}_{i=1}^{\infty}$ are piecewise disjoint. Then next step is to show that $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} A_i^\star$. First, note that since $A_i^\star \subset A_i$ for all $i \in \mathbb{N}$. Then $\cup_{i=1}^{\infty} A_i^\star \subset \cup_{i=1}^{\infty} A_i$. Now, we show that $\cup_{i=1}^{\infty} A_i \subset \cup_{i=1}^{\infty} A_i^\star$. Suppose $x \in \cup_{i=1}^{\infty} A_i$. Then, let i_0 be the smallest index integer such that $x \in A_{i_0}$. In particular, $x \notin A_1, x \notin A_2, \ldots, x \notin A_{i_0-1}$. Then $x \in A_{i_0}^\star \subset \cup_{i=1}^{\infty} A_i^\star$. Therefore, $\cup_{i=1}^{\infty} A_i^\star \subset \cup_{i=1}^{\infty} A_i^\star$. Then $\cup_{i=1}^{\infty} A_i^\star = \cup_{i=1}^{\infty} A_i$. $P(\cup_{i=1}^{\infty} A_i) = P(\cup_{i=1}^{\infty} A_i^\star) = \sum_{i=1}^{\infty} P(A_i^\star) \leq \sum_{i=1}^{\infty} P(A_i)$, since $A_i^\star \subset A_i$.

Corollary 1.1. *Let $A_1, A_2, \ldots, A_n$ be events. Then $P(\cap_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$. Proof: $P(\cap_{i=1}^{n} A_i) = 1 - P((\cap_{i=1}^{n} A_i)^c) = 1 - P(\cup_{i=1}^{n} A_i^c) \geq 1 - \sum_{i=1}^{n} P(A_i^c) = 1 - (n - \sum_{i=1}^{n} P(A_i)) = \sum_{i=1}^{n} P(A_i) - (n-1)$, since $P(\cup_{i=1}^{n} A_i^c) \leq \sum_{i=1}^{n} P(A_i^c)$.*

Problem 1. *Suppose $\{A_i\}_{i=1}^{\infty}$ are events such that $P(A_i) = 1$ for all $i \in \mathbb{N}$ such that $P(\cap_{i=1}^{\infty} A_i) = 1$.*[1]

Independent Events: Events A and B are said to be independent if $P(A \cap B) = P(A)P(B)$. (...) Finite sample spaces and equally likely outcomes

Example. *We roll a dce twice (tolls are assumed to be independent) and we write down the number obtained at each roll. The sample space $\Omega = \{(a,b), a, b \in \{1,2,3,4,5,6\}\}$. $|\Omega| = 36$. Since the two rolls are independent, the probability of $\{(a,b)\}$ is $1/6x1/6 = \frac{1}{36}$. The outcomes are equally likely. Let $\Omega = \{w_1, \ldots, w_n\}$ be a finite sample space with $|\Omega| = N$ we say that the outcomes $w_1, \ldots, w_N$ are equally likely if $P(\{w_1\}) = P(\{w_2\}) = \ldots = P(\{w_N\}) = \frac{1}{N}$. Let $A \subset \Omega$. Since Ω is finite, then A is finite and $A = \{w_1, w_2, \ldots, 2_d\}$ and $|A| = d$, $w_1, \ldots, w_n$ are the distinct d elements of A. $P(A) = \sum_{i=1}^{d} P(\{w_i\}) = \frac{1}{N} + \frac{1}{N} + \ldots + \frac{1}{N} = \frac{d}{N} = \frac{|A|}{|\Omega|}$.*

Example. *A fair coin is flipped twice. Suppose the flips are independent. The sample space $\Omega = \{HH, HT, TH, TT\}$ outcomes in Ω are equally likely with probability $1/4$.*

1. *Probability of getting at least one head, call this event A. Find P(A): $A = \{HH, HT, TH\}$, $|A| = 3$, $P(A) = \frac{3}{4}$.*

Counting

Theorem 2 (Fundamental Theorem of Counting)**.** *With m elements $a_1, \ldots, a_m$ and n elements $b_1, \ldots, b_n$, one can form $m \times n$ distinct pairs (a,b) where $a \in \{a_1, \ldots, a_m\}$ and $b \in \{b_1, \ldots, b_n\}$.*

[1] He'll prolly ask this on an exam.

Example. *how many 7 digit telephone numbers can one form? 10 chosen for first, 10 chosen for second, ..., 10 chosen for 7th How many 7-digit telephone numbers can one form if the third digit is even: 10 chosen for first, 10 chosen for second, 5 chosen for third, 10 chosen for fourth, ..., 10 chosen for seventh.*

Permutations

Example. *In a class of 100 students, 2 students are chosen. The first one wins 100\$ and the second one wins 75\$. How many choices are there if (1) a student can win twice (sampling with replacement) (2) a student can win at most once (sampling without replacement). (1) 100 choices for first, 100 choices for second, so* 100^2 *choices. (2) 100 choices for first, 99 choices for second, so* 100×99 *choices.*

Permutations: An ordered arrangement of r distinct objects is called a permutation. The number of ways of randomly choosing r ordered objects form n objects without replacement is denoted by P_r^n and is equal to $\frac{n!}{(n-r)!}$. In total, the number of ordering is $n \times (n-1) \times \ldots \times (n-r+1)$. $P_r^n = n \times (n-1) \times \ldots \times (n-r+1) = \frac{n!}{(n-r)!}$.

Example. *Two students are chosen to form a committee. In here many ways can the choice be done? 100 choices for first one, 99 choices for second one, 2! ways of ordering the two students, so* $\frac{100 \times 99}{2!}$ *ways.*

3 Appendix

4 Solutions

5 Useful Links